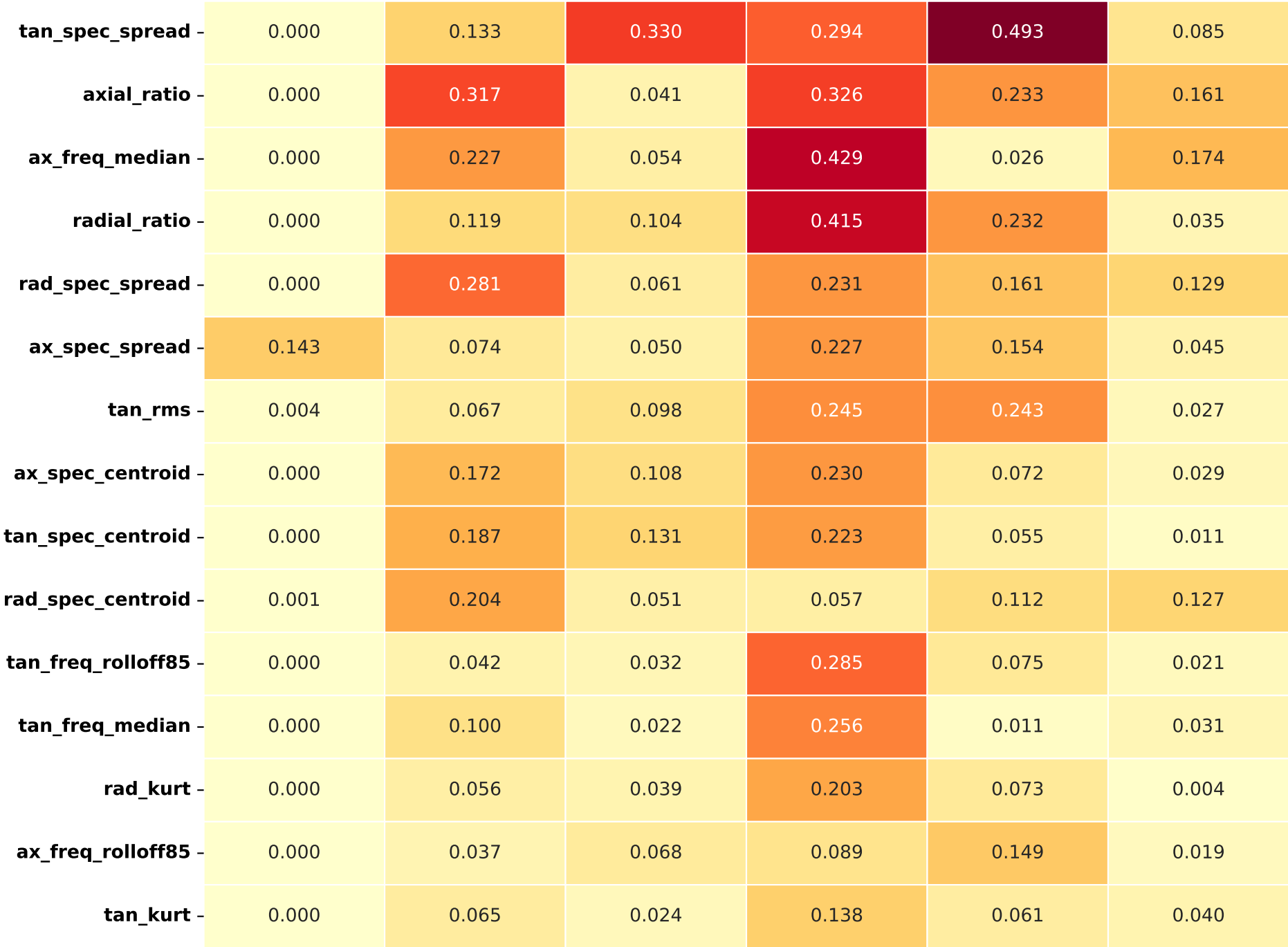


Physics-Aligned Feature Discriminative Power
Permutation Importance on Full Test Set

Feature (Top 15)



Physics Validation:

- Spectral spread dominates (top 6 features)
→ Harmonic-rich signatures from MaFaulDa coupling
- Radial features > axial (rad_spec_spread #5)
→ Flexible coupling transmits forces radially
- Kurtosis secondary (rad_kurt #13)
→ Mild bearing faults (6-35g) produce moderate impulsiveness

Fault Class